

Exploratory Data Analysis Report

E-Commerce Sales Dataset | 1,200 Records | 14 Columns

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1. Problem Statement

This analysis investigates an e-commerce sales dataset to understand customer purchasing behavior, identify revenue trends, and surface actionable business insights. The core question: *Which factors drive total purchase value, and where are the key risks and opportunities in this sales data?*

2. Methodology

Tools used: Python, Pandas, Matplotlib. The analysis followed a structured EDA pipeline:

- Descriptive statistics to understand central tendency and spread
- Trend analysis of monthly revenue over a 2.5-year period (Jan 2023 – Jun 2025)
- Distribution analysis via histogram to assess data shape and skewness
- Outlier detection using the IQR (Interquartile Range) method with box plot visualization
- Correlation analysis across all numerical variables using the Pearson coefficient
- Categorical breakdown of Product, OrderStatus, PaymentMethod, and ReferralSource

3. Statistical Analysis

3.1 Descriptive Statistics

The table below summarizes key statistics for the four numerical columns:

Statistic	Quantity	Unit Price	Items in Cart	Total Price
Mean	2.95	\$356.41	5.49	\$1,053.97
Median	3.00	\$364.21	5.00	\$823.62
Std Dev	1.41	\$197.18	2.28	\$819.86
Min	1	\$11.39	1	\$11.39
Max	5	\$699.93	10	\$3,456.40

3.2 Interpretation

The mean TotalPrice (\$1,053.97) is significantly higher than the median (\$823.62), a gap of \$230. This divergence signals right-skewed data: a small number of high-value transactions pull the mean upward. For business reporting, the median is the more reliable measure of a typical customer transaction. UnitPrice shows the highest variability (Std Dev \$197.18), reflecting a broad product price range from low-cost items (\$11.39) to premium products (\$699.93). TotalPrice variability (Std Dev \$819.86) is substantial, confirming wide variation in customer spending.

4. Trend Analysis

Monthly sales were computed by grouping TotalPrice by calendar month across the full dataset (Jan 2023 – Jun 2025).



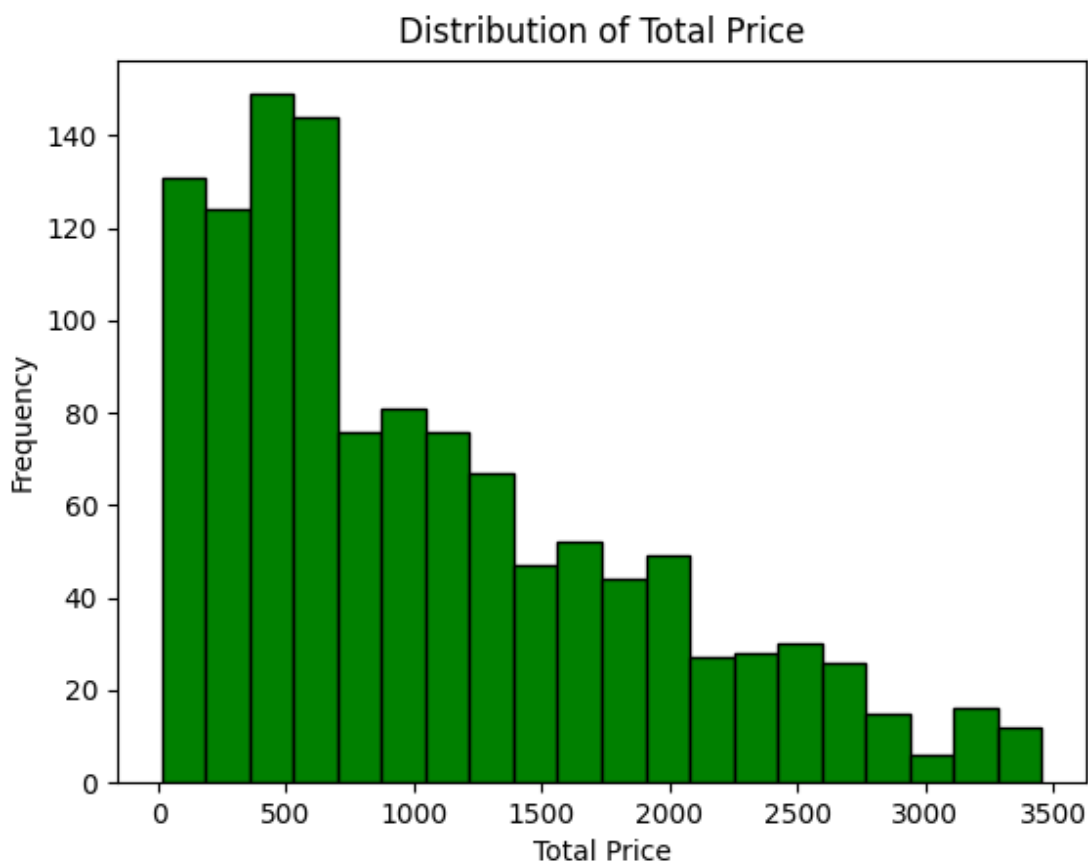
Key findings:

- **June 2024 — \$68,069** (the highest single month in the dataset)
- **April 2023 — \$27,752** (145% below the peak)
- Notable surge in May–June 2023 (\$63,837) and again in June 2024 (\$68,069), suggesting a recurring mid-year demand pattern

- 2025 shows a recovery trend: Q1 was slower (\$29K–\$39K range) but May–June 2025 is climbing back toward \$43K–\$53K
- Sales declined sharply after June 2024 through late 2024 a gap worth investigating (promotions ending, seasonality, or external factors)

Business diagnosis: The mid-year peaks (May–June) across multiple years suggest a seasonal pattern. The business should prepare inventory and marketing campaigns ahead of this window each year.

5. Distribution Analysis



A histogram of TotalPrice values reveals a right-skewed distribution (skewness = 0.891). Most transactions are clustered in the \$0–\$1,000 range, with frequency declining sharply at higher values. This is a classic long-tail distribution common in e-commerce datasets.

Skewness values for all numeric columns:

- Quantity: 0.028 nearly symmetric, purchases are evenly distributed 1–5 units
- UnitPrice: -0.027 slight negative skew, prices are fairly balanced
- ItemsInCart: 0.001 perfectly symmetric

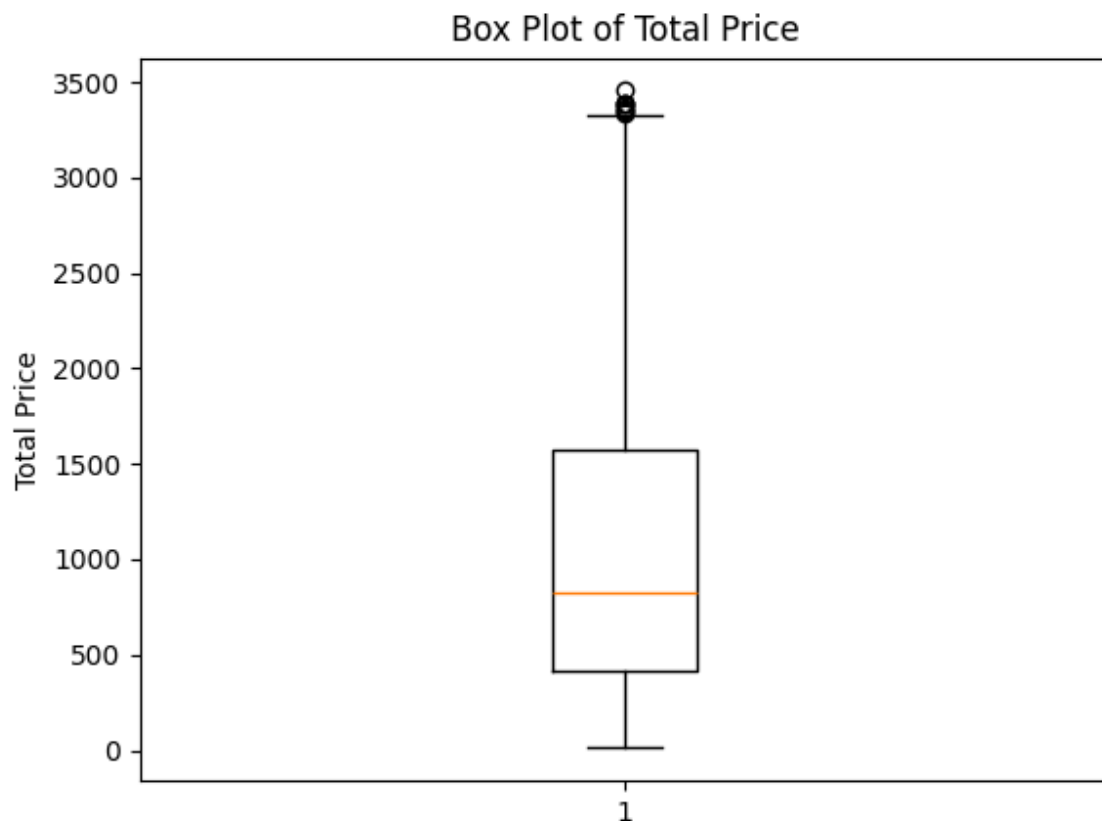
- TotalPrice: 0.891 significant right skew; a few large orders dominate the upper tail

Business diagnosis: Standard marketing based on average spend (\$1,054) will misrepresent most customers, who typically spend around \$824 (median). Pricing strategies and loyalty tiers should be anchored to the median, not the mean.

6. Outlier Detection

Outliers were identified using the IQR method:

- $Q1 = \$410.52$ | $Q3 = \$1,578.48$ | $IQR = \$1,167.96$
- Lower fence = $-1,341.41$ (no negative prices exist, no low-end outliers)
- Upper fence = $\$3,330.41$
- Outliers detected: 8 transactions above the upper fence

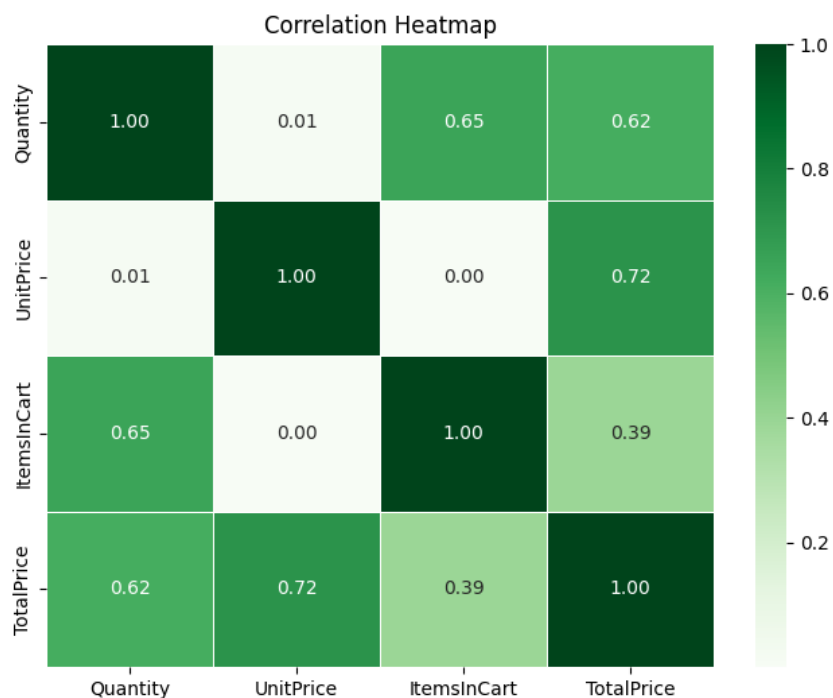


The 8 outlier values range from \$3,334.00 to \$3,456.40, all clustered tightly near the upper fence. This pattern suggests these are genuine high-value orders (signal), not data entry errors (noise). They represent VIP or bulk purchases and should be investigated, not removed. No lower-bound outliers exist, confirming data quality on the low end.

7. Correlation Analysis

The Pearson correlation matrix for all numerical variables:

	Quantity	Unit Price	Items in Cart	Total Price
Quantity	1.000	0.015	0.650	0.615
Unit Price	0.015	1.000	0.001	0.717
Items in Cart	0.650	0.001	1.000	0.393
Total Price	0.615	0.717	0.393	1.000



Key findings:

- **UnitPrice-TotalPrice ($r = 0.717$)** the price of the product is the biggest driver of total spend

- **Quantity-TotalPrice ($r = 0.615$)** buying more units also meaningfully increases total spend
- **Quantity- ItemsInCart ($r = 0.650$)** customers who buy more of one item also tend to have fuller carts
- **Weak:** UnitPrice has almost no relationship with Quantity ($r = 0.015$) or ItemsInCart ($r = 0.001$)

8. Categorical Analysis

The dataset contains four categorical columns that reveal critical business context beyond numerical trends.

8.1 Order Status

- Cancelled: 250 orders (20.8%) -- *highest category*
- Returned: 247 orders (20.6%)
- Pending: 237 orders (19.8%)
- Shipped: 235 orders (19.6%)
- Delivered: 231 orders (19.2%)

Business diagnosis: Only 19.2% of orders are successfully delivered. Combined cancellations and returns account for 41.4% of all orders a critical red flag. This level of non-completion suggests fulfillment, product, or customer expectation issues that need urgent investigation.

8.2 Referral Source

- Instagram: 259 orders (21.6%) -- *top channel*
- Email: 250 orders (20.8%)
- Google: 241 orders (20.1%)
- Facebook: 228 orders (19.0%)
- Referral: 222 orders (18.5%)

Instagram is the top acquisition channel. Marketing budget should be weighted toward Instagram and Email, which together account for 42.4% of orders.

8.3 Product Revenue

- Chair: \$195,620 | Printer: \$195,613 (virtually tied at the top)
- Laptop: \$192,127 | Tablet: \$186,569
- Monitor: \$175,651 | Desk: \$167,460 | Phone: \$151,722

Revenue is relatively balanced across products, suggesting no single product dependency. Chair and Printer are slight leaders worth prioritizing for promotions and inventory.

8.4 Payment Methods

All five methods (Online, Cash, Credit Card, Debit Card, Gift Card) are evenly distributed (approximately 20% each), indicating no payment friction for customers

9. Key Observations

- TotalPrice is right-skewed (skewness = 0.891): the median (\$823.62) is a better business metric than the mean (\$1,053.97)
- June is consistently the peak month of sales across 2023 and 2024 a reliable seasonal pattern to plan around
- UnitPrice is the strongest driver of TotalPrice ($r = 0.717$), suggesting premium product sales are the key revenue lever
- 8 high-value outliers (\$3,334–\$3,456) appear to be genuine bulk/VIP orders, not errors a segment worth nurturing
- 41.4% of orders are cancelled or returned, the most critical business risk identified in this dataset
- Instagram is the top acquisition channel; Email follows closely as the most cost-effective digital channel
- Chair and Printer are the top revenue-generating products, each contributing ~\$195K total

10. Conclusion & Recommendations

This EDA successfully identified trends, distributions, outliers, correlations, and categorical patterns in 1,200 e-commerce transactions. Three priority recommendations based on the data:

- Investigate the 41.4% cancellation/return rate immediately this is the largest single threat to revenue identified in the analysis
- Launch mid-year promotions in May–June to capitalize on the seasonal demand peak observed in 2023 and 2024
- Focus marketing spend on Instagram and Email (top two acquisition channels) and prioritize premium-priced products (Chairs, Printers, Laptops) which drive the highest TotalPrice values